

L-22: Characteristic Functions

Introduction: Complex Numbers and Vector Space

Consider the mapping:

$$z = a + ib \longleftrightarrow \begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$$

where $\mathbb{R}^2 = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \mid a, b \in \mathbb{R} \right\}$.

We define binary operations on $\mathbb{R}^2$ to mirror the logic of $\mathbb{C}$:

- **Addition:** $\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$
- **Multiplication:** Defined such that it mimics $(a + ib)(c + id)$:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac - bd \\ bc + ad \end{pmatrix}$$

By defining the imaginary unit vector as $i = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, we find that $i \cdot i = \begin{pmatrix} -1 \\ 0 \end{pmatrix} = -1$.

Geometric Interpretation and Euler's Formula

The Complex Plane

The modulus of z corresponds to the Euclidean norm:

$$|z| = \sqrt{a^2 + b^2}$$

In polar coordinates (r, θ) , a point on the unit circle ($|z| = 1$) is given by:

$$z = (\cos \theta, \sin \theta)$$

$$|z| \leq 1, Z \in \mathbb{C}$$

$$z_{\text{in}} = (r \cos \theta, r \sin \theta)$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Derivation of Euler's Formula

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots = \sum_{m=0}^{\infty} (-1)^m \frac{\theta^{2m}}{(2m)!}$$

$$+ i \sin \theta = i \left[\frac{\theta}{1!} - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right] = i \sum_{m=1}^{\infty} (-1)^{m-1} \frac{\theta^{2m-1}}{(2m-1)!}$$

$$\cos \theta + i \sin \theta = 1 + \frac{i\theta}{1!} + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!}$$

$$= e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Properties:

- Multiplication: $e^{i(\theta+\psi)} = e^{i\theta} \cdot e^{i\psi}$
- Euler's Identity: $e^{i\pi} + 1 = 0$
- Modulus: $|e^{i\theta}| = 1$

Characteristic Functions (Definition)

Let X be a random variable with PDF (or PMF) f_X . The **Characteristic Function** of X , denoted $\phi_X(t)$, is a function $\phi : \mathbb{R} \rightarrow \mathbb{C}$ defined as:

$$\phi_X(t) = \mathbb{E}[e^{itX}]$$

Using Euler's formula, this can be written as:

$$\phi_X(t) = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)]$$

Properties

1. $\phi_X(t)$ is always defined.
2. $\phi_X(0) = 1$.
3. $|\phi_X(t)| \leq 1$.

$$\overbrace{\left| \mathbb{E}[e^{itX}] \right|}^{\text{Norm}} \leq \mathbb{E} \left[\underbrace{|e^{itX}|}_{\text{Norm}=1} \right] = 1$$

4. If X & Y are independent:

$$\begin{aligned} \phi_{X+Y}(t) &= \mathbb{E}[e^{it(X+Y)}] \\ &= \mathbb{E}[e^{itX} \cdot e^{itY}] \\ &= \mathbb{E}[e^{itX}] \mathbb{E}[e^{itY}] \\ &= \phi_X(t) \cdot \phi_Y(t) \end{aligned}$$

Examples of Characteristic Functions**1. Constant Random Variable**

Let $X = c$ (constant).

$$\phi_X(t) = \mathbb{E}[e^{itc}] = e^{itc} = \cos(tc) + i \sin(tc)$$

2. Bernoulli(p)

Values: 0, 1 with probabilities $q = 1 - p$ and p .

$$\begin{aligned} \phi_X(t) &= \mathbb{E}[e^{itX}] = q \cdot e^{it(0)} + p \cdot e^{it(1)} \\ \phi_X(t) &= q + pe^{it} = (1 - p) + p(\cos t + i \sin t) \end{aligned}$$

3. Geometric(p)

Let $X \sim \text{Geo}(p)$ with support $\{0, 1, 2, \dots\}$. $P(X = n) = q^n p$ where $q = 1 - p$.

$$\phi_X(t) = \sum_{n=0}^{\infty} e^{itn} q^n p = p \sum_{n=0}^{\infty} (qe^{it})^n$$

Using the geometric series sum $\frac{1}{1-r}$:

$$\phi_X(t) = \frac{p}{1 - qe^{it}}$$

Separating into Real and Imaginary parts:

$$\begin{aligned} \phi_X(t) &= \frac{p}{1 - q(\cos t + i \sin t)} = \frac{p}{(1 - q \cos t) - iq \sin t} \\ &= \frac{p[(1 - q \cos t) + iq \sin t]}{(1 - q \cos t)^2 + q^2 \sin^2 t} \end{aligned}$$

$$\sum_{n=0}^{\infty} \cos(nt) q^n = \frac{1 - q \cos t}{(1 - q \cos t)^2 + q^2 \sin^2 t}$$

$$\sum_{n=0}^{\infty} \sin(nt) q^n = \frac{q \sin t}{(1 - q \cos t)^2 + q^2 \sin^2 t}$$

Normal Distribution Characteristic Function

Standard Normal $X \sim N(0, 1)$

The characteristic function is given by:

$$\phi_Z(t) = \mathbb{E}[e^{itZ}] = \int_{-\infty}^{\infty} e^{itz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

Combine the exponents: $-\frac{1}{2}(z^2 - 2itz)$. Completing the square:

$$z^2 - 2itz = (z - it)^2 - (it)^2 = (z - it)^2 + t^2$$

Substituting back into the integral:

$$\begin{aligned} \phi_Z(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}((z-it)^2 + t^2)} dz \\ &= e^{-t^2/2} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-it)^2} dz}_{=1} \end{aligned}$$

The integral is 1 because it represents the area under a "shifted" complex normal density (contour integration proof required for rigor, but assumed 1 here).

$$\phi_Z(t) = e^{-t^2/2}$$

General Normal $X \sim N(\mu, \sigma^2)$

Using the linear transformation property: $X = \mu + \sigma Z$.

$$\phi_X(t) = \mathbb{E}[e^{it(\mu + \sigma Z)}] = e^{it\mu} \phi_Z(\sigma t)$$

$$\phi_X(t) = e^{it\mu - \frac{1}{2}\sigma^2 t^2}$$

Alternative Derivation: Series Expansion for $N(0, 1)$

$$\begin{aligned} \phi_Z(t) &= \mathbb{E}[\cos(tZ)] + i\mathbb{E}[\sin(tZ)] \\ &= \mathbb{E}\left[\sum_{n=0}^{\infty} (-1)^n \frac{(tZ)^{2n}}{(2n)!}\right] + 0 \quad (\text{Odd moments are 0}) \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!} \mathbb{E}[Z^{2n}] \end{aligned}$$

Recall even moments of standard normal: $\mathbb{E}[Z^{2n}] = \frac{(2n)!}{2^n n!}$.

$$= \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!} \frac{(2n)!}{2^n n!} = \sum_{n=0}^{\infty} \frac{(-t^2/2)^n}{n!} = e^{-t^2/2}$$